

Tax Revenue and Economic Growth in Shanghai: A
VECM-Based Empirical Analysis
上海市税收收入与经济增长关系研究——基于 VECM 模
型的实证分析

a thesis

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Abstract

This study adopts the annual data of Shanghai from 1990 to 2023, constructs a vector error correction model (VECM) to examine the dynamic relationship between tax revenue and economic growth. The results show that $\ln GDP$ and $\ln TAX$ are both integrated of order one, with a unique long-run cointegrating relationship, and the long-run output elasticity is about 0.768; economic output bears the burden of adjustment, returning to equilibrium at a rate of about 14.97% per year, and tax revenue shows weak exogeneity. The impulse response shows that there is an obvious time delay asymmetry in the transmission of the two: the positive effect of tax expansion on economic growth appears about two years later and continues without attenuating, with a permanent effect; while the driving effect of economic growth on tax revenue is immediate and continuous. Variance decomposition further confirms this feature. Research shows that Shanghai's tax revenue and economic growth have formed a stable two-way feedback relationship dominated by the economy, which provides long-run empirical evidence for the formulation of local tax policies for core cities.

本研究采用上海市 1990-2023 年的年度数据，构建向量误差修正模型（VECM），探究税收收入与经济增长之间的动态关系。研究结果表明： $\ln GDP$ 与 $\ln TAX$ 均为一阶单整序列，二者存在唯一的长期协整关系，长期产出弹性约为 0.768；经济产出承担了主要的调整负担，以每年约 14.97% 的速率回归均衡状态，而税收收入表现出弱外生性。脉冲响应分析显示，二者的传导过程存在显著的时间延迟不对称性：税收扩张对经济增长的正向效应在约两年后显现，且持续存在未出现衰减，具有永久性影响；经济增长对税收收入的拉动

效应则是即时且持续的。方差分解进一步验证了这一特征。研究表明，上海市税收收入与经济增长已形成以经济为主导的稳定双向反馈关系，这为核心城市地方税收政策的制定提供了长期经验证据。

Keywords: tax revenue; economic growth; cointegration; vector error correction model(VECM)

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Chapter 1

Introduction

The dynamic interaction between tax level and economic growth has always been the core issue in the fields of macroeconomics and finance. Moderate taxation is important for the country to cope with market failures, investment in infrastructure and the provision of basic public services such as education and medical care. In some countries, when the macro tax burden reaches the threshold level of Gross Domestic Product (GDP), it can greatly promote the inclusive growth of the economy (Choudhary et al., 2024). However, adjustments in tax policies may also alter the behavior and decision-making of enterprises and investors by influencing capital costs and market expectations. Existing research has found that an increase in taxes may raise the cost of capital use and curb corporate investment, while also causing certain distortions to real economic variables and asset prices (Ghilardi & Zilberman, 2025). Focusing on the Chinese context, against the backdrop of rapid economic transformation, domestic scholars have conducted more systematic and in-depth research on the economic effects of taxation. Early nationwide time series empirical studies indicated that, over a long period after the reform and opening up (1978-2007), the shock of the tax revenue share had an overall positive effect on macroeconomic growth (Liu, 2009). However, as China's economy shifts from a stage of high-speed growth to a stage of high-quality development, the relationship between taxation and the economy has become more complex. For example, based on panel data from 31 provinces nationwide covering 2010–2022, Zhang et al. (2024) point out that there is obvious regional heterogeneity in the impact of overall tax revenue on economic growth. This means that the conclusion based on the national historical average may not accurately portray the economic reality of a core region. Therefore, it is of

great significance to conduct further exploration of the relationship between taxation and economic growth at the regional level. Existing studies mostly focus on the national or provincial level, leaving relatively limited coverage in the long-run time series dynamic research on China's economic transformation and the leading cities of opening up. As a highly developed core city in China's economy and finance sector, Shanghai has a different ability to absorb tax shocks compared to traditional inland provinces, due to its huge size and industrial structure. Therefore, this study aims to deeply analyze the correlation between tax revenue and macroeconomic output in Shanghai from 1990 to 2023. Specifically, this paper aims to answer the following two core questions: Over a 34-year time span, has the tax scale in Shanghai established a long-run stable cointegration relationship with economic growth? What are the dynamic transmission characteristics between the tax scale of Shanghai and the macroeconomy? To answer these questions, this paper selects the annual macroeconomic data of Shanghai from 1990 to 2023, establishes a vector error correction model (VECM) based on the long-run cointegration relationship between variables, and uses impulse response functions and variance decomposition to depict the dynamic transmission characteristics of tax shocks on economic growth. The empirical results of this study not only help to reveal the long-run equilibrium relationship and dynamic adjustment characteristics between the tax scale of big cities and economic growth, but also provide long-term time series empirical references for Shanghai to formulate more targeted tax policies in the current economic transformation period.

Chapter2

Literature Review

In international research, a large amount of theoretical and empirical literature has been accumulated regarding the relationship between taxation and economic growth. From a theoretical perspective, Besley and Persson (2014), in a study based on cross-national historical data, pointed out that the capacity for tax extraction is a core pillar of modern state capacity. An effective tax system and market institutions, as well as the construction of the rule of law, form a complementary evolutionary relationship, and are necessary conditions for supporting long-run economic growth; low tax capacity will cause developing countries to fall into a vicious cycle of poverty, while a stable fiscal foundation provides institutional guarantees for the country to continuously invest in public goods. Stoilova and Patonov (2013) further pointed out from the mechanism level that taxation can raise funds for public expenditures, undertake the function of income redistribution, and have an impact on the allocation of resources in the economy, all of which are necessary prerequisites for achieving economic growth. This theoretical judgment has also been further verified in a large number of cross-national empirical studies. An International Monetary Fund (IMF) study by Gaspar et al. (2016) , based on panel data from 139 countries from 1965 to 2011, confirmed that when a country's macro tax burden crosses the critical threshold of approximately 12.75% of GDP, it will trigger a substantial leap in tax compliance norms and the country's fiscal capacity, thereby promoting the sustained acceleration of per capita GDP. The latest research by Choudhary et al. (2024) further supports this logic. When a country's tax-to-GDP ratio exceeds the critical threshold, it will create significant fiscal space for developing countries in productive public expenditure areas such as health and education, thereby significantly

driving economic growth in the medium and long term. Empirical studies focusing on a single country or a specific region also have numerous findings supporting the positive stimulating effect of taxation on economic growth. A study on multiple African countries from 2004 to 2013 found that tax revenue was significantly positively correlated with GDP, and taxation had a positive promoting effect on the economic growth of African countries (Onakoya et al., 2017). Tsaurai (2021) selected the panel data of 11 emerging market countries from 2008 to 2018 and used estimation methods such as dynamic Generalized Method of Moments (GMM) and random effects to reach the conclusion that tax revenue had a significant positive effect on the economic growth of emerging markets, indicating that the positive incentive effect of taxation is not limited to developed economies. At the time series level of a single country, Islam (2019) used the data of Bangladesh from 2001 to 2017 as a sample, and through the classical linear regression model and Johansen cointegration test, verified that tax revenue had a positive relationship with economic growth, and there was a long-run stable correlation between the two. Studies that further applied the VECM revealed more dynamic transmission characteristics: Agunbiade and Idebi (2020), based on the annual data of Nigeria from 1981 to 2019, found that the core tax revenue had a long-run stable cointegration relationship with actual GDP, and indirect taxes had the most prominent positive stimulating effect on economic growth, with its variance contribution accumulating to 80.05% in the 10th period. This finding further indicates that by lowering the analytical perspective to the long-run sequence test of a single country can reveal the tax structure differences that cross-national panel studies cannot capture. However, taxes can also have a constraining impact on economic growth. From a theoretical perspective, taxes reduce the disposable resources of the private sector, resulting in a crowding-out effect on private investment. When this crowding-out effect exceeds the productive benefits brought by tax conversion into public services, the long-run economic growth rate will decline. This

negative effect caused by tax distortion has been demonstrated in a series of empirical samples from specific countries and regions. At the regional large-sample level, based on the panel data from 9 member states of the Association of Southeast Asian Nations (ASEAN), Nguyen and Darsono (2022) confirmed that tax revenue has a significant negative impact on the per capita GDP growth rate at the linear level. More importantly, when the research perspective is lowered to the long-run sequential test of a single economy, this negative mechanism can also be precisely captured. Dladla and Khobai (2018) used the autoregressive distributed lag (ARDL) model to analyze South Africa and found that income tax and actual GDP show a robust negative correlation in the short term and long term; while Collaku et al. (2023) applied a VECM to data from Kosovo and indicated that the fluctuations in tax scale had a negative effect on macro output in the long term. Several studies jointly suggest that the negative effect of taxes may be more evident in economies with relatively weak institutional environments and industrial foundations. The empirical divergence of this tax growth effect showing both positive and negative trends cannot be simply attributed to methodological differences. Instead, it points to a more fundamental fact: The actual impact of taxation on economic growth will exhibit systematic differences depending on the tax system structure, stage of development, and sample characteristics.

Turning the research focus back to China, empirical studies on the relationship between taxation and economic growth also achieved certain results. However, the research conclusions vary due to differences in the dependent variable setting, method selection, and sample scope. At the national level, most existing studies generally support a long-run positive correlation between taxation and economic growth. The VECM study based on national time series data indicated that there was a stable long-run cointegration relationship between tax revenue and GDP, and the growth rate of tax revenue was higher than that of GDP in the long term (Wang & Wang, 2024); earlier vector autoregressive model (VAR)

study also found that the shock to the proportion of tax revenue had a continuous positive impact on the GDP growth rate, and the positive effect reached its peak in the fourth year after the shock (Liu, 2009). This time lag characteristic provides an important early reference for subsequent examination of the transmission rhythm of tax policies. However, this nationwide positive consensus does not imply the consistency of tax economic relations across different regions. China has a vast territory, and different regions have objective differences in natural resources, human capital, and industrial structure, which leads to significant spatial heterogeneity in the economic effects of taxation. A study based on panel data from 31 provinces confirmed that moderate tax extraction had a positive incentive effect on economic growth, but this positive incentive effect showed significant differences in the eastern, central, western, and northeastern regions (Zhang et al., 2024). This heterogeneity was more prominent in province-specific studies. Yi and Suyono (2014), using data from Hebei Province from 1978 to 2011 as a sample, combined with the modified tax multiplier effect theory and the polynomial distributed lag model (PDL), found that current taxes had a positive impact on GDP, but the negative impact of taxes had a significant lag feature. The taxes in the first and third lags had negative effects on economic growth, and this effect became more obvious over time; while a VAR study using data from Anhui Province from 1994 to 2015, with tax revenue and GDP levels as the core variables, found that there was a significant long-run positive cointegration relationship between the two, with a long-run elasticity coefficient of approximately 0.75 (Chen, 2018). The conclusions of the two provinces profoundly revealed the high degree of situational dependence on tax and economic relations at the provincial level in China, and also indicated that any research conclusions based on national data or a single province cannot accurately portray the actual transmission mechanisms of other regions.

Overall, our review of the domestic and foreign literature mentioned above identified two

aspects that the current research could benefit from further exploration in terms of sample scope and method selection. From the sample level, most of the existing literature relied on transnational samples or national inter-provincial panel data, focusing on identifying generalizable average patterns, ignoring the significant differences in industrial structure, financial efficiency, and development stage in specific regions to a certain extent; although there had been some accumulation of provincial-level studies, conclusions differed significantly across regions. Research focusing on the long-run dynamic transmission between tax revenue and economic growth in Shanghai, a core Chinese city, remained limited. Methodologically, early studies used VAR models to depict the short-term impulse responses among variables, often ignoring the possible long-run stability relationship between non-stationary time series. If the analytical framework lacked the portrayal of potential long-run constraints, it might miss the cointegration relationship, which leads to biased statistical inference. This study takes Shanghai as its core case and constructs an annual time-series dataset covering 1990 to 2023. It systematically examines the long-run equilibrium relationship and dynamic transmission patterns between local tax revenue and economic output. In terms of the econometric strategy, this study adopts the VECM, incorporating long-run steady-state tests and short-term dynamic adjustment into the same analytical framework to make up for the shortcomings of purely short-term shock analysis, thereby providing more solid empirical evidence for optimizing tax policies in core cities and supporting macro growth.

Chapter 3

Methods

3.1 Variable Selection and Definition

In order to deeply analyze the dynamic interaction between Shanghai's tax revenue and macroeconomic growth, this section will detail the core variables, data sources and data preprocessing steps used in empirical analysis.

Gross Domestic Product (GDP): The regional gross domestic product of Shanghai for the current year is selected as the variable for measuring the aggregate economic output and economic growth level of Shanghai. This indicator can comprehensively reflect the overall scale and development trend of economic activity in Shanghai during a specific period. The data unit used in this article is "billion yuan".

Tax Revenue (TAX): The total tax revenue of Shanghai is selected as the variable to measure the local fiscal strength and the scale of government tax collection. This indicator reflects the scale of social wealth that the Shanghai municipal government mobilizes through taxation. The unit is also "billion yuan".

3.2 Data Sources and Preprocessing

This study selects the annual time series data of Shanghai from 1990 to 2023 as the research sample. The original absolute values of Shanghai's regional gross domestic product and total tax revenue used in the article were all obtained from the official database of the *Shanghai Statistical Yearbook* spanning the period from 1990 to 2023. The data series within the sample period is continuous and complete, without any outliers or missing values, ensuring

the authority and accuracy of the data. During the data preprocessing stage, since macroeconomic time series data usually exhibit heteroscedasticity and the absolute numerical values are large, this paper uniformly applied logarithmic processing to the original values of GDP and TAX before conducting the empirical tests. This step helped alleviate the interference of heteroscedasticity on model estimation accuracy and enabled the estimated parameters in the subsequent VECM to directly reflect the "elasticity" relationship between variables. These operations generated new time series variables, denoted as lnGDP and lnTAX.

3.3 ADF Unit Root Test

Before modeling the time series data, it is necessary to test the stationarity of the series. This study employs the Augmented Dickey-Fuller test to determine the order of integration of lnGDP and lnTAX. The ADF test controls for serial correlation in the residuals by including lagged difference terms in the regression equation. Given that Shanghai's GDP and TAX both exhibit a clear upward trend over the sample period, this paper adopts a specification with both a constant term and a time trend. The standard regression model is specified as follows:

$$\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \sum_{i=1}^k \gamma_i \Delta y_{t-i} + \varepsilon_t$$

Where:

y_t is the time series variable to be tested, and Δ is the first-difference operator, $\Delta y_t = y_t - y_{t-1}$;

α is the constant term, and βt is the time trend term;

k is the optimal lag length determined to eliminate residual autocorrelation ;

ε_t is the independent and identically distributed white noise error term.

Test Hypotheses and Decision Criteria:

Null Hypothesis(H_0): $\delta = 0$, indicating that the series contains a unit root and is non-stationary.

Alternative Hypothesis (H_1): $\delta < 0$, indicating that the series does not contain a unit root and is stationary.

In practice, if the calculated ADF statistic is less than the critical value at a specific significance level, the null hypothesis is rejected, and the series is considered stationary.

Conversely, if it is not rejected, the series requires further differencing until it becomes stationary. If a variable's first difference is stationary, the variable is $I(1)$, that is, integrated of order one.

3.4 Vector Autoregression (VAR) Model

In order to accurately capture the dynamic interaction effect between the tax scale of Shanghai and the macroeconomic growth, this paper first constructs an unconstrained VAR model as a preliminary analytical framework, thereby laying a systematic foundation for subsequent cointegration tests and VECM. Different from traditional static regression models with their limitations in dealing with endogeneity issues, the VAR model does not need to pre-determine the endogeneity and exogeneity of each variable in the system. Instead, all variables are regarded as endogenous variables, and their lagged terms are used as the right-side explanatory variables. Based on the two log-transformed time series variables selected in this study, $\ln(GDP)_t$ and $\ln(TAX)_t$, the unrestricted bivariate VAR(p) model is specified as follows:

$$Y_t = C + \sum_{i=1}^p \Phi_i Y_{t-i} + \varepsilon_t$$

Expanding this equation for the specific variables of this article, its two-dimensional matrix form is:

$$\begin{bmatrix} \ln (GDP)_t \\ \ln (TAX)_t \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \sum_{i=1}^p \begin{bmatrix} \phi_{11}^{(i)} & \phi_{12}^{(i)} \\ \phi_{21}^{(i)} & \phi_{22}^{(i)} \end{bmatrix} \begin{bmatrix} \ln (GDP)_{t-i} \\ \ln (TAX)_{t-i} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

Where:

$\mathbf{Y}_t = [\ln (GDP)_t, \ln (TAX)_t]^T$ is a 2×1 column vector of endogenous variables, representing the economic aggregate and tax scale in period t ;

$\mathbf{C} = [C_1, C_2]^T$ is a 2×1 vector of constant terms;

p is the optimal lag order (or lag length) of the VAR model;

Φ_i is a 2×2 autoregressive coefficient matrix to be estimated for the i -th lag. Its elements capture the lagged marginal effects of the variables on themselves as well as the cross-variable delayed impacts;

ε_t is a 2×1 vector of random error terms.

By constructing this baseline VAR model and determining the optimal lag order, this study can proceed to formally test for long-run cointegrating relationships between the variables.

3.5 Johansen Cointegration Test

The logical premise of the cointegration test is that the related time series must be integrated of the same order. After determining the stationarity of the variables through the preceding ADF test, if the variables are confirmed to be integrated of the same order, our study will further determine whether there is a long-run equilibrium relationship between the two. This paper adopts the Johansen cointegration test. This method can simultaneously test all possible cointegration vectors in the system, not only effectively alleviating the endogeneity interference, but also having higher statistical power and robustness. The basic logic of the Johansen test is to transform an unrestricted VAR model with an optimal lag order of p through differencing into the following vector error correction form:

$$\Delta Y_t = \alpha + \beta t + \Pi Y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta Y_{t-i} + \varepsilon_t$$

Among them, the rank of the matrix Π determines the number of cointegration relationships among the variables. In the specific empirical operation, this paper mainly uses the trace statistic test to determine the cointegration rank r . The calculation formula of the trace statistic is as follows:

$$\lambda_{trace}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i)$$

Where:

T is the effective sample size;

$\hat{\lambda}_i$ represents the i th estimated eigenvalue obtained through the eigenvalue decomposition;

k represents the number of endogenous variables in the system.

Test Hypotheses and Decision Rule:

The trace test follows a sequential testing logic starting from $r = 0$ (indicating the absence of a cointegration relationship), and incrementing r step by step.

Null Hypothesis (H_0): The number of cointegration vectors is less than or equal to r

Alternative Hypothesis (H_1): The number of cointegration vectors is greater than r

If the calculated trace statistic is greater than the critical value at the corresponding significance level, the null hypothesis is rejected, indicating that there are more cointegration relationships in the system. If the final test result shows that there is a long-run cointegrating trajectory between the tax revenue scale of Shanghai and economic growth, this study will base on this and further construct a VECM model with an error correction term.

3.6 Specification of the Vector Error Correction Model (VECM)

If the Johansen cointegration test confirms a long-run cointegrating relationship between $\ln GDP$ and $\ln TAX$, estimating a simple unrestricted VAR model would omit long-run constraint information, leading to specification bias. Therefore, an error correction term (ECM) must be introduced to construct a VECM. The economic rationale of the VECM is that it not only captures short-run dynamic shocks among variables through lagged differenced terms, but also accounts for the mean-reverting adjustment mechanism when the system deviates from long-run equilibrium through the error correction term. Based on the bivariate system in this study, the specific matrix expansion of the standard VECM(p-1) model is as follows:

$$\begin{bmatrix} \Delta \ln GDP_t \\ \Delta \ln TAX_t \end{bmatrix} = ECM_{t-1} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \sum_{i=1}^{p-1} \begin{bmatrix} \gamma_{11,i} & \gamma_{12,i} \\ \gamma_{21,i} & \gamma_{22,i} \end{bmatrix} \begin{bmatrix} \Delta \ln GDP_{t-i} \\ \Delta \ln TAX_{t-i} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

Where the long-run error correction term ECM_{t-1} is defined as:

$$ECM_{t-1} = \beta_1 \ln GDP_{t-1} + \beta_2 \ln TAX_{t-1} + C$$

Where:

Δ is the first-difference operator, representing short-run fluctuations of the variables;

ECM_{t-1} is the one-period lagged error correction term, reflecting the magnitude of deviation from the long-run equilibrium trajectory between tax revenue scale and economic growth in the previous period;

$\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$ means adjustment coefficient vector. If the system has a mechanism for converging toward equilibrium, the elements of this vector are expected to be negative and statistically significant. It measures the speed and strength with which the system returns to long-run equilibrium after a deviation caused by external shocks;

$\begin{bmatrix} \gamma_{11,i} & \gamma_{12,i} \\ \gamma_{21,i} & \gamma_{22,i} \end{bmatrix}$ is the 2×2 short-run dynamic coefficient matrix for the i-th lag, measuring the

lagged marginal impact of each variable's short-run fluctuations on the current period;

$\begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$ is a random disturbance term.

3.7 Diagnostic Tests

To verify the rationality of the model setup, after the VECM estimation is completed, this paper will conduct two diagnostic tests on the system residuals. The Portmanteau test is used to examine whether there is serial correlation in the residuals, in order to verify the appropriateness of the lag order selection; the multivariate Autoregressive Conditional Heteroskedasticity (ARCH) test is used to examine whether there is conditional heteroskedasticity in the residuals in order to confirm the reliability of the model estimation results.

3.8 Impulse Response Function

Although the VECM model can calculate the error correction mechanism among variables, the economic meanings of its numerous short-term lag parameters are not intuitive enough. Therefore, this paper uses the impulse response function to capture the dynamic response of the system to external shocks. The mathematical essence of the impulse response is to transform the VECM system into an infinite-order vector moving average (VMA) form. The basic mathematical expression is as follows:

$$Y_t = \sum_{i=0}^{\infty} \Theta_i \varepsilon_{t-i}$$

Where:

Y_t is the vector of endogenous variables;

ε_{t-i} is the error term vector of the system in the $t-i$ period;

Θ_i is the impulse response coefficient matrix, which depicts the marginal response of the variables in the future period i after the system receives a one standard deviation shock.

By drawing the impulse response diagram based on the above formula, this paper can visually observe the lag length, direction of effect (positive pull or negative crowding out), and duration of the impact on regional economic growth caused by the change in the scale of tax revenue in Shanghai.

3.9 Variance Decomposition

Variance decomposition is mainly used to assess the relative contribution of each shock source to the overall fluctuations of the system. To overcome the information overlap caused by the possible contemporaneous correlation among disturbance terms, this paper adopts the Cholesky decomposition method to orthogonalize the covariance matrix of the disturbance terms in the empirical study. Variance decomposition calculates the prediction error variance of the system for the next h steps and decomposes it according to the causes into the variance proportions caused by orthogonalized shocks of $\ln GDP$ and $\ln TAX$, thereby defining the relative importance of different variables in the evolution of the system. It is particularly important to note that since the Cholesky decomposition result is highly sensitive to the order of variable ordering, a reasonable variable arrangement is the core prerequisite for ensuring the economic significance of the empirical results. Following the underlying logic of macroeconomic operation and fiscal policy transmission: the fluctuations of the macroeconomy have a direct decisive effect on the tax base and can be transformed into a contemporaneous response of tax revenue within a relatively short collection cycle; while tax policy adjustments affect the real economy, and from the implementation of the policy to the substantive changes in investment and production decisions by enterprises, there is often a long transmission lag. Therefore, in the orthogonalization process, this paper sets the

economic aggregate $\ln GDP$ as the first variable and the $\ln TAX$ as the second variable to ensure that the decomposition trajectory conforms to the common sense of real macroeconomic operation.

Chapter 4

Results

4.1 Descriptive Statistics and Time Series Trend Analysis

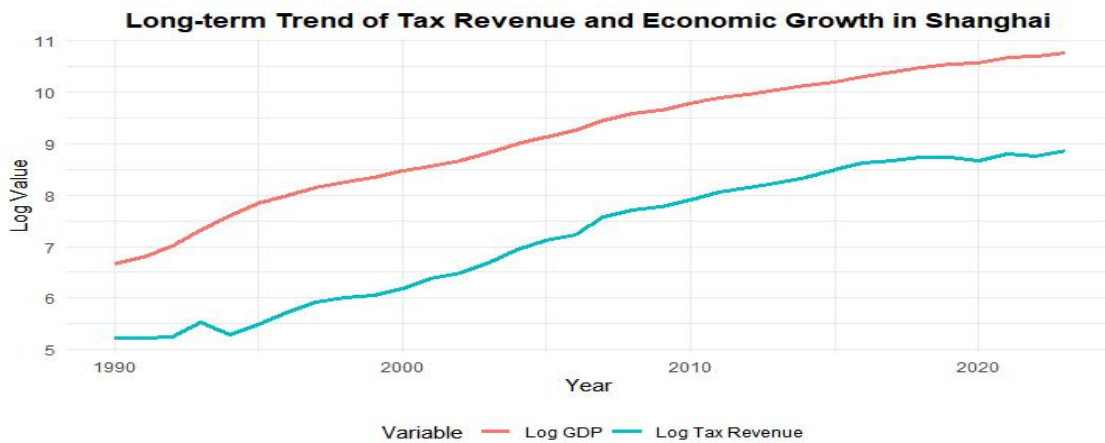
Table 4.1 presents the descriptive statistics of the core variables $\ln\text{GDP}$ and $\ln\text{TAX}$. The sample period is from 1990 to 2023, covering 34 annual observations. The results show that the standard deviations of the variables are 1.229 and 1.304 respectively, and the corresponding coefficients of variation are 13.4% and 18.1% respectively. This indicates that the dispersion of the data series within the observation period is relatively low, without the disturbance of extreme outliers, and the overall data quality is good, providing a solid data foundation for subsequent time series modeling.

Table 4. 1: Descriptive Statistics of Core Variables

Variable	N	Mean	Std. Dev.	Min	Max	CV
$\ln\text{GDP}$	34	9.151	1.229	6.661	10.763	0.134
$\ln\text{TAX}$	34	7.201	1.304	5.209	8.869	0.181

Furthermore, Figure 4.1 depicts the long-run evolution trend of $\ln\text{GDP}$ and $\ln\text{TAX}$ in Shanghai. The results show that both variables exhibited a continuous upward trend during the sample period, without showing any significant mean reversion characteristics.

Figure 4. 1: The trend charts of lnGDP and lnTAX



4.2 Sequence Stationarity Test

From the test results in Table 4.2, the ADF statistics of the original series of lnGDP and lnTAX are -2.588 and 0.105 respectively. Both of them fail to reject the null hypothesis of the existence of unit roots at the 10% significance level, indicating that the original series of Shanghai's economic volume and tax revenue are both non-stationary sequences. To eliminate the non-stationarity of the sequences, this paper further performs first-order differencing on the above two variables. The test results show that the null hypothesis is rejected for $\Delta \ln \text{GDP}$ at the 10% level and for $\Delta \ln \text{TAX}$ at the 5% level. In conclusion, lnGDP and lnTAX are both first-order integrated sequences, that is, $I(1)$ sequences, meeting the preconditions for constructing the VECM model and conducting cointegration tests.

Table 4. 2: ADF Unit Root Test Results

Variable	ADF Statistic	1% Critical Value	5% Critical Value	10% Critical Value	Result
lnGDP	-2.588	-4.15	-3.50	-3.18	Non-stationary
lnTAX	0.105	-4.15	-3.50	-3.18	Non-stationary
$\Delta \ln \text{GDP}$	-2.8*	-3.58	-2.93	-2.60	Stationary
$\Delta \ln \text{TAX}$	-3.449**	-3.58	-2.93	-2.60	Stationary

4.3 Determination of the Optimal Lag Order

After confirming that both variables are $I(1)$ series, considering the sensitivity of Johansen cointegration test to the setting of model lag order, this paper first constructs an unconstrained VAR model to determine the optimal lag order. According to the lag order determination criterion, the smaller the value of the information index, the better trade-off between the goodness of fit and parameter simplicity of the model. Table 4.3 shows that, at lag order one, the core statistical criteria of AIC, SC, HQ, and FPE all reach the minimum simultaneously. Based on the consistency of multiple criteria, this paper determines the optimal lag order for the basic VAR model to be 1, and uses this as the reference for conducting the Johansen cointegration test subsequently.

Table 4. 3: Selection Criteria for Optimal VAR Model Lag Length

Lag length (p)	AIC	SC	HQ	FPE
1	-12.597*	-12.223*	-12.477*	3.39e-06*
2	-12.508	-11.947	-12.328	3.74e-06
3	-12.298	-11.551	-12.059	4.68e-06
4	-12.212	-11.278	-11.913	5.24e-06

Note: * indicates the minimum value of the respective information criterion.

4.4 Johansen Cointegration Test Results

Based on the previous confirmation that all variables are of first-order integrated sequences, this study further examines whether there exists a long-run stable equilibrium relationship between $\ln GDP$ and $\ln TAX$. Considering that the Johansen cointegration test relies on the VECM framework, and in this framework, the lag order of the difference terms is equal to the lag order of the original VAR model minus one. If the 1st-order lag selected based on the information criterion in the previous text is directly used, it will result in the short-term

dynamic adjustment term of VECM being reduced to zero order, which means that the model will be unable to effectively capture the short-term fluctuation characteristics of the variables, and is prone to cause autocorrelation problems in the residuals, thereby violating the premise of the Johansen test. In view of this, in order to more fully absorb the short-term dynamic shocks in the sequence to ensure that the residuals satisfy the white noise assumption, while taking into account the principle of parameter simplicity to avoid excessive consumption of the limited sample degrees of freedom, and considering that the differences between the information criterion values and the optimal values of the VAR lag of 2nd order are extremely small, this paper expands the VAR lag order for the cointegration test to the 2nd order ($K=2$). The specific test results are shown in Table 4.4. From Table 4.4, it can be seen that at the 1% significance level, the trace test results significantly reject the null hypothesis of no cointegration relationship ($r=0$), but fail to reject the hypothesis of at most one cointegration relationship ($r \leq 1$).

Table 4. 4 : Johansen Trace Test Results

Null Hypothesis	Trace Statistic	1% Critical Value	Conclusion
$r=0$ (No cointegration)	41.27	24.60	Reject
$r \leq 1$ (At most one)	12.51	12.97	Fail to reject

The above results provide strong evidence that within the sample period from 1990 to 2023, there exists a unique and extremely significant long-run cointegration relationship between the economic growth of Shanghai and its tax revenue. This indicates that although both may fluctuate in the short term due to various exogenous shocks, they possess a stable equilibrium attribute in the long run, meeting the prerequisite conditions for constructing an error correction model. Based on the normalized cointegration vector, the long-run equilibrium equation of the two can be expressed as:

$$\ln GDP_t = 0.768 \ln TAX_t + 4.004$$

The cointegration equation shows that Shanghai's tax revenue has a significant long-run positive effect on macroeconomic growth. For every 1% increase in tax revenue, economic output increases in the long term by 0.768%. This suggests a stable proportional relationship between the expansion of Shanghai's tax scale and the economic fundamentals. Tax revenue has played an extremely important foundational role in supporting and reflecting the stable operation of the regional economy.

4.5 VECM Results

The estimation results of the VECM in this paper are shown in Table 4.5. The estimation results of the error correction term reveal the long-run adjustment mechanism of the system. In the $\Delta \ln GDP_t$ equation, the coefficient of the error correction term is -0.1497, which is significantly negative at the 1% significance level, indicating that when the long-run equilibrium relationship between the tax scale and the economic total in Shanghai deviates, the economic total will return to the equilibrium direction at a rate of approximately 14.97% per year. The long-run equilibrium has a strong self-repairing ability. In contrast, the coefficient of the error correction term in the $\Delta \ln TAX_t$ equation is 0.0428, with a p-value of 0.708, and fails to pass the significance test, indicating that tax revenue does not have a significant error correction to the long-run equilibrium deviation, indicating weak exogeneity of tax revenue in this dynamic system. Further observation of the transmission effect of short-term dynamic fluctuations, in the difference equation of $\Delta \ln GDP_t$, the coefficient of the first-order lagged difference term of $\ln GDP$ is 0.4382, and it is significantly positive at the 1% level, indicating that the economic growth in Shanghai has a strong short-term inertia and continuity; while the coefficient of the first-order lagged difference term of $\ln TAX$ is 0.0312, which fails to pass the significance test ($p = 0.629$), indicating that the short-term fluctuations in the tax scale do not have a significant impact on economic growth. At the same time, in the

$\Delta \ln \text{TAX}_t$ equation, the coefficient of the first-order lagged difference term of $\ln \text{GDP}$ is 0.9056 ($p = 0.075$), which is marginally significant at the 10% level, indicating that economic growth has a certain short-term positive leading effect on tax revenue; but the coefficient of the short-term lag term of tax itself is not significant, indicating that tax revenue lacks significant autoregressive inertia in the short term. In summary, the economic total of Shanghai plays a dominant role in both the long-run equilibrium repair and the short-term dynamic transmission, and tax revenue shows a passive following characteristic of growth.

Table 4. 5: Vector Error Correction Model (VECM) Estimation Results

Variable	$\Delta \ln \text{GDP}_t$	$\Delta \ln \text{TAX}_t$
ECT_{t-1}	-0.1497*** (0.000)	0.0428 (0.708)
$\Delta \ln \text{GDP}_{t-1}$	0.4382*** (0.004)	0.9056* (0.075)
$\Delta \ln \text{TAX}_{t-1}$	0.0312 (0.629)	-0.0477 (0.835)

Note: Values in parentheses are p-values.

4.6 Model Diagnostics and Residual Tests

Before conducting the dynamic impact analysis, it is necessary to perform residual diagnosis on the estimated model to ensure the validity of the subsequent impulse response and variance decomposition results. The results show that the p-value obtained from the Portmanteau test is 0.884, which is significantly greater than the significance level of 0.05. Therefore, the null hypothesis of no serial correlation cannot be rejected. At the same time, the p-value obtained from the multivariate ARCH test is 0.881, indicating that there is no significant conditional heteroscedasticity (ARCH effect) in the residual sequence. These findings confirm that the residuals follow a white noise process.

4.7 Impulse Response Analysis

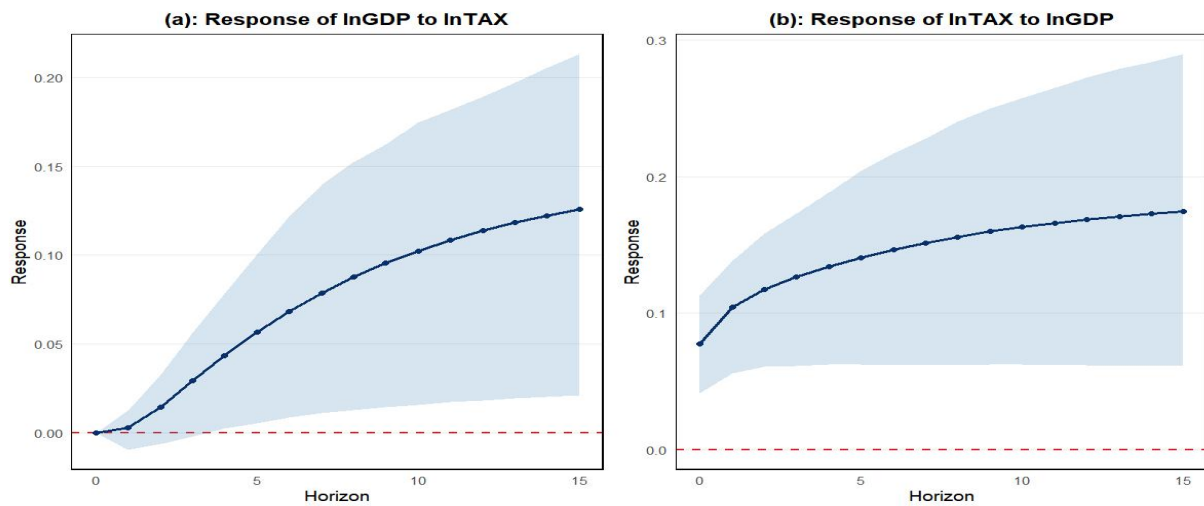
Figure 4.2(a) shows the response trajectory of the economic total after a one-unit standard deviation positive shock to the tax scale. $\ln GDP$ exhibits a brief lag at the beginning of the shock and then shows a steady and continuous upward trend. Further investigation into its convergence characteristics reveals that this response curve does not exhibit an attenuation trend towards a zero mean, but rather presents a typical permanent shock pattern. The cumulative positive effects continue to accumulate as the number of forecast periods increases. This indicates that the exogenous expansion of the tax scale does not merely cause short-term economic disturbances, but can permanently raise the long-run equilibrium baseline of the economic total. Considering the evolution path of the 95% confidence interval, although the response failed to pass the significance test in the first two periods, starting from the third period, the lower limit of the confidence interval has completely crossed the zero axis and has remained in the positive range. This result provides strong empirical support, indicating that in the macroeconomic operation framework of Shanghai, the expansion of the tax scale undergoes a short-term lagged transmission and will have a significant and persistent positive effect on economic growth.

Figure 4.2(b) presents the dynamic response trajectory of tax revenue after the economic total is subjected to a positive shock. The response solid line shows a positive value starting from the current period (period 0) and continues to increase monotonically as the lag period increases, without showing any signs of decay or decline throughout the entire prediction window. The upper and lower bounds of the 95% confidence interval remain strictly above the zero axis throughout the entire prediction window, indicating that this positive response is statistically significant throughout. This indicates that the expansion of the economic total can immediately and continuously drive the growth of tax revenue, and the driving effect does not need to undergo time lag transmission, and continuously accumulates and strengthens over

time.

By combining the response characteristics of the two figures and confirming them with the previous VECM estimation results, an important rule can be found: The dynamic timing rhythm of the tax scale and economic growth in the transmission shows asymmetry. Specifically, the positive pull of the tax scale expansion on the economic total requires about 2 periods of transmission lag before it can have a substantive impact; while the driving effect of economic growth on tax revenue is immediate and continuous throughout.

Figure 4. 2: Impulse Response Analysis



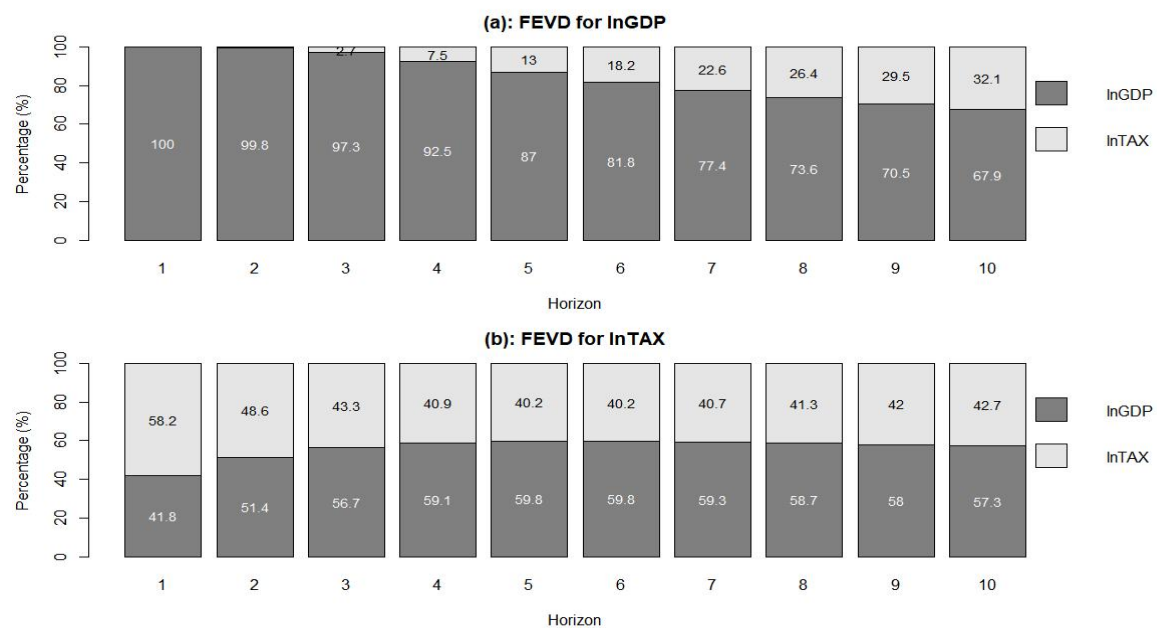
4.8 Variance Decomposition

Based on the variance decomposition results for the next 10 periods shown in Figure 4.3, it can be observed that the interaction between Shanghai's tax revenue and its economic output exhibits a significant asymmetric characteristic.

Looking at the Figure 4.3(a), in the early stages (periods 1 to 2), the fluctuations in economic output are almost entirely explained by its own shocks. However, as the lag period increases, the variance contribution rate of tax revenue to economic output shows a steady upward trend. By the 5th period, this contribution rate has risen to 13.0%; by the end of the 10th period, it

further climbs to 32.1%. Combined with the impulse response results mentioned earlier, this indicates that although there is a policy transmission lag in the short term, in the medium and long-run macroeconomic operation, the change in tax revenue constitutes an important exogenous driving force for the long-run equilibrium increase of Shanghai's economic output. From Figure 4.3(b), it can be seen that, in contrast to the lagged effect of tax revenue on economic output, the variance contribution of economic output to tax revenue shows a very strong instantaneous triggering and leading characteristic. In the 1st period of the shock, the explanatory power of economic output for the fluctuation of tax revenue immediately reaches 41.8%; subsequently, this proportion expands rapidly and remains at a peak level of nearly 60% (59.8%) from the 5th to the 6th period. After that, although this proportion slightly declines, it still remains at 57.3% by the 10th period. This result provides strong empirical evidence, indicating that the change in Shanghai's tax revenue is highly endogenous to the expansion of the macroeconomy; the economic fundamentals not only immediately broaden the current tax base but also are the core determining variable for the long-run trend of tax revenue.

Figure 4. 3: Results of Forecast Error Variance Decomposition



Chapter 5

Discussion

This article aims to systematically summarize and deeply analyze the empirical findings regarding the bidirectional dynamic relationship between tax revenue and regional economic output in Shanghai from 1990 to 2023. Based on the test results of the VECM, it is indicated that there exists a long-run stable positive cointegration relationship between tax revenue and regional economic output in Shanghai; a significant error correction mechanism is systematically presented, with the adjustment speed approximately at 14.97% per year, while tax revenue exhibits a relatively weak exogeneity. The impulse response function shows that the positive stimulating effect of tax scale on economic growth does not decay in the short term after being impacted, but remains stable at a significantly positive level within a 15-period window, presenting the characteristic of a permanent positive effect. The results of variance decomposition also support this judgment: by the 10th period, the variance contribution of tax revenue to the total economic fluctuation reached 32.1%, while the variance contribution of the total economic volume to the tax fluctuation reached 57.3%, and the two showed an obvious asymmetric structure in terms of transmission rhythm and contribution intensity.

Several findings in the empirical results of this paper are worth discussing and comparing with existing literature. This article identifies a stable long-run cointegration relationship between Shanghai's TAX and GDP, with a long-run elasticity coefficient of 0.768. This conclusion corroborates the findings of Chen (2018). He built a VAR model based on the data of tax revenue and regional gross domestic product from 1994 to 2015 in Anhui Province, and also found that there is a long-run stable cointegration relationship between the two; for

every 1% change in tax revenue, GDP changes by 0.75% in the same direction. Both studies have detected a stable cointegration relationship in two different provincial samples, and the long-run elasticity magnitudes are very close. This consistency is not accidental but reflects that, during the relatively long period after the tax-sharing system reform in China, there is indeed a stable long-run equilibrium proportion relationship between local tax revenue and economic output.

On a more macro level, the conclusion of this paper is consistent with Liu's (2009) study based on a VAR model covering the period from 1978 to 2007. Both studies indicate that taxation has a positive impact on the economic total. It should be noted that Liu (2009) selected the relative quantitative indicator of the proportion of tax revenue in GDP, focusing on the impulse effect of the macro tax burden on the economic growth rate; this paper takes the absolute scale of tax revenue as the core variable and examines its long-run equilibrium impact on the total economic output. There are essential differences in the variables between the two, so there is no direct numerical comparability, but the consistency in direction provides robustness verification for the conclusion of this paper. However, the findings of this paper differ from some studies. Yi and Suyono (2014), based on data from Hebei Province (1978–2011), modified the traditional tax multiplier theory and constructed a 4th-order PDL model. They found that current taxation can drive GDP growth, but the negative effect has a lag and gradually strengthens over time. Using the latent threshold time-varying parameter factor-augmented vector autoregressive (LT-TVP-FAVAR) model to analyze data from China from 2007 to 2023, Zhao et al. (2024) indicated that an increase in total taxation has an inhibitory effect on economic growth, but this effect has weakened after the new normal. The findings of these studies contrast significantly with the continuous positive cumulative effect of tax shocks presented in the Shanghai sample of this paper.

Understanding this divergence requires returning to the dual perspectives of the objective

differences in China's economic operation and the choice of methodology. From the perspective of regional structure, Zhang et al. (2024), using panel data from 31 provinces in China, confirmed that the positive incentive effect of tax revenue on economic growth has significant regional heterogeneity. Shanghai, with a focus on high-value-added modern service and financial industries, has a relatively limited distortion effect of tax burden on enterprise production decisions, and tax revenue can be efficiently converted into public services and infrastructure supporting high-end industries; Hebei, with heavy industry and traditional manufacturing as the mainstay, has a higher marginal sensitivity of enterprises to tax burden, and the crowding-out effect gradually emerges in the lag period. From the perspective of methodology, this paper adopts the VECM framework to explicitly identify the long-run equilibrium relationship between non-stationary variables through error correction terms, and to depict the regression speed after deviation from equilibrium. Yi and Suyono (2014) adopted a PDL model in log-level form, estimating the multi-period marginal effect of tax shock in a single equation form, but did not explicitly handle the non-stationarity of variables and long-run decomposition. The LT-TVP-FAVAR model used by Zhao et al. (2024), in a stabilized multivariate VAR framework, uses time-varying parameters to characterize the dynamic transmission differences of tax shock in different economic system stages, with the advantage of capturing structural changes rather than long-run equilibrium. The three studies correspond to different methodological emphases, which complement each other rather than directly competing on the research questions.

The academic significance of this study lies in the long-run empirical investigation of the relationship between local taxation and regional economic growth, taking Shanghai as the starting point. Existing studies mostly take the national total or provincial panel as the analysis unit, and the national total data is difficult to reflect the structural differences between regions. Based on the 34-year cycle data of Shanghai, this article fully covers key

institutional nodes such as tax reform and economic "new normal". This kind of observation sample with institutional span can provide an appropriate empirical basis for systematically examining the long-run equilibrium relationship between taxation and economic growth. On this basis, this article further reveals the bidirectional asymmetric transmission of their interaction: the impact of economic growth on tax revenue is immediate and strong, while the effect of tax revenue on economic growth exhibits a lagged and lasting effect. This understanding helps to deepen the theoretical understanding of the relationship between local taxation and economy, and also provides an empirical basis for grasping the transmission rhythm and timing of tax policy in policy formulation. The high-intensity and permanent positive pull characteristics shown in Shanghai not only provide specific empirical cases for subsequent comparable research in other economically developed cities, but also provide a reference empirical starting point for similar studies.

There are still some problems that need to be further deepened in this study: the existing empirical framework is not deep enough for the internal logical analysis of taxation affecting the economy. Although this study confirms the continuous driving effect of tax scale on economic growth from the macro total level, because the model only includes total indicators, the model fails to jointly test the specific financial expenditure structure (such as people's livelihood investment) or the investment decision-making of micro-subjects. This total-based analysis framework makes it difficult to clearly identify the specific channels through which taxation has led to economic growth. Future research can consider introducing structural data such as tax details or fiscal expenditure classification, or combining the intermediary effect model with the structural vector autoregression model (SVAR) to more accurately test the actual transmission vein of tax policies. At the level of measurement methods, the VECM model adopted in this paper is based on linear and parameter constant assumptions, and it is difficult to capture the nonlinear structural mutations that may be caused by multiple changes

in the fiscal and tax system during the sample period of up to 34 years. In view of the above shortcomings, future research can consider the introduction of a time-varying parameter vector autoregression model (TVP-VAR) or the Markov regime-switching VAR model (MS-VAR). This kind of nonlinear measurement framework can effectively relax the assumption of constant parameters, so as to more accurately identify the time-varying characteristics and structural transformation laws of tax policy effects under different economic cycles and reform stages.

Chapter 6

Conclusion

This article examines the relationship between tax revenue and total economic volume based on the annual data of Shanghai from 1990 to 2023. The research results show that there is a long-run stable cointegration relationship between the two. The long-run output elasticity is approximately 0.768. The long-run positive impact of taxation on economic growth is cumulative, and there is no downward trend. In terms of the transmission rhythm, the effect of taxation on economic growth exhibits a lag of approximately two years, while the impact of economic growth on taxation is almost instantaneous. This provides specific insights for understanding the logic of the long-run interaction between taxation and economy in developed regions of China. The research results also provide certain insights for the formulation and evaluation of tax policies. The lag characteristic of tax shocks means that policy effects take time to manifest. Evaluating the effect of tax policies based only on current year or the next year's data is likely to underestimate their actual contribution. Policy evaluation should leave a longer observation period.

Different regions in China vary greatly in terms of economic foundation, industrial structure and financial system. A unified policy framework may not be suitable for all regions. A long-run careful investigation of representative core cities such as Shanghai is conducive to supplementing the local experience that is difficult to present in

national and provincial research.

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Data and Code Availability

The data used in this study are from the Shanghai Statistical Yearbook (Shanghai Municipal Bureau of Statistics), publicly available at

<https://tjj.sh.gov.cn/tjnj/20250331/9f8ec62cc2234485b0aa411b8d967c37.html>.

The replication code is available at

<https://github.com/wensiruizhang/shanghai-tax-gdp-vecm>